

two examples, the calculations for finding the probabilities of two of the possible F_2 genotypes ($YYRR$ and $YyRR$) are shown below:

$$\text{Probability of } YYRR = \frac{1}{4} (\text{probability of } YY) \times \frac{1}{4} (RR) = \frac{1}{16}$$

$$\text{Probability of } YyRR = \frac{1}{2} (Yy) \times \frac{1}{4} (RR) = \frac{1}{8}$$

The $YYRR$ genotype corresponds to the upper left box in the larger Punnett square in Figure 14.8 (one box = $\frac{1}{16}$). Looking closely at the larger Punnett square in Figure 14.8, you will see that 2 of the 16 boxes ($\frac{1}{8}$) correspond to the $YyRR$ genotype.

Now let's see how we can combine the multiplication and addition rules to solve even more complex problems in Mendelian genetics. Imagine a cross of two pea varieties in which we track the inheritance of three characters. Let's cross a trihybrid with purple flowers and yellow round seeds (heterozygous for all three genes) with a plant with purple flowers and green wrinkled seeds (heterozygous for flower color but homozygous recessive for the other two characters). Using Mendelian symbols, our cross is $PpYyRr \times Ppyyrr$. What fraction of offspring from this cross are predicted to exhibit the recessive phenotypes for *at least two* of the three characters?

To answer this question, we can start by listing all genotypes we could get that fulfill this condition: $ppyyRr$, $ppYyrr$, $Ppyyrr$, $PPyyrr$, and $ppyyrr$. (Because the condition is *at least two* recessive traits, it includes the last genotype, which shows all three recessive traits.) Next, we calculate the probability for each of these genotypes resulting from our $PpYyRr \times Ppyyrr$ cross by multiplying together the individual probabilities for the allele pairs, just as we did in our dihybrid example. Note that in a cross involving heterozygous and homozygous allele pairs (for example, $Yy \times yy$), the probability of heterozygous (Yy) offspring is $\frac{1}{2}$ and the probability of homozygous (in this case, yy) offspring is $\frac{1}{2}$. Finally, we use the addition rule to add the probabilities for all the different genotypes that fulfill the condition of at least two recessive traits resulting from our $PpYyRr \times Ppyyrr$ cross, as shown below:

$ppyyRr$	$\frac{1}{4} (\text{probability of } pp) \times \frac{1}{2} (yy) \times \frac{1}{2} (Rr) = \frac{1}{16}$
$ppYyrr$	$\frac{1}{4} (pp) \times \frac{1}{2} (Yy) \times \frac{1}{2} (rr) = \frac{1}{16}$
$Ppyyrr$	$\frac{1}{2} (Pp) \times \frac{1}{2} (yy) \times \frac{1}{2} (rr) = \frac{2}{16}$
$PPyyrr$	$\frac{1}{4} (PP) \times \frac{1}{2} (yy) \times \frac{1}{2} (rr) = \frac{1}{16}$
$ppyyrr$	$\frac{1}{4} (pp) \times \frac{1}{2} (yy) \times \frac{1}{2} (rr) = \frac{1}{16}$
Chance of <i>at least two</i> recessive traits	$= \frac{6}{16} = \frac{3}{8}$

In time, you'll be able to solve genetics problems faster by using the rules of probability than by filling in Punnett squares.

We cannot predict with certainty the exact numbers of progeny of different genotypes resulting from a genetic cross. But the rules of probability give us the *likelihood* of various outcomes. Usually, the larger the sample size, the closer the results will conform to our predictions. Mendel understood

this statistical feature of inheritance and had a keen sense of the rules of chance. It was for this reason that he set up his experiments so as to generate, and then count, large numbers of offspring from his crosses.

CONCEPT CHECK 14.2

1. For any gene with a dominant allele A and recessive allele a , what proportions of the offspring from an $AA \times Aa$ cross are expected to be homozygous dominant, homozygous recessive, and heterozygous?
2. Two organisms, with genotypes $BbDD$ and $BbDd$, are mated. Assuming independent assortment of the B/b and D/d genes, write the genotypes of all possible offspring from this cross and use the rules of probability to calculate the chance of each genotype occurring.
3. **WHAT IF?** Three characters (flower color, seed color, and pod shape) are considered in a cross between two pea plants: $PpYyli \times ppYyli$. Using the rules of probability, determine the fraction of offspring predicted to be homozygous recessive for at least two of the three characters.

For suggested answers, see Appendix A.

CONCEPT 14.3

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

In the 1900s, geneticists extended Mendelian principles not only to diverse organisms, but also to patterns of inheritance more complex than those described by Mendel. For the work that led to his two laws of inheritance, Mendel chose pea plant characters that turn out to have a relatively simple genetic basis: Each character is determined by one gene, for which there are only two alleles, one completely dominant and the other completely recessive. (There is one exception: Mendel's pod shape character is actually determined by two genes.) Not all heritable characters are determined so simply, and the relationship between genotype and phenotype is rarely so straightforward. Mendel himself realized that he could not explain the more complicated patterns he observed in crosses involving other pea characters or other plant species. This does not diminish the utility of Mendelian genetics, however, because the basic principles of segregation and independent assortment apply even to more complex patterns of inheritance. In this section, we will extend Mendelian genetics to hereditary patterns that were not reported by Mendel.

Extending Mendelian Genetics for a Single Gene

The inheritance of characters determined by a single gene deviates from simple Mendelian patterns when alleles are not completely dominant or recessive, when a particular gene has